

Default DataFolder should point to the « diff » subfolder

Data Browser

Current Data Folder: root:NO3CalData:diff:

Display

- ☒ Waves
- ☒ Variables
- ☒ Strings
- ☒ Info
- ☒ Plot

New Data Folder

Save Copy

Browse Expt...

Delete

Execute Cmd...

Filter

Wave: cpc_counts
Type: Single Float 32 bit
Rows: 32 Start: 0 Delta: 1 Units:
Data Units:
Size: 752 bytes

Waveform plot showing a peak around 12:00:00.

TOPWERK v3.3.0

File info User data Log Plot Workflows Misc

ACSM data processing part 2

☐ Capture Vaporizer?

☒ Airbeam correction, diagnos. and Hz unit conversion

Results and general plotting options

ACSM data and plotting options

- ☐ Use ug/m3 units (else ions/s)
- ☐ Use the airbeam correction
- ☐ Use the CDCE (if generated)

☒ New plot ☐ Append to top plot

Data set

☒ Difference ☐ Sample ☐ Filter

ACSM species and fragments time series plots

Species: NO₃, NO₂+

Time base

☒ Native ☐ Avg. Hrs: 1 ☐ Diurnal: 1 hrs

☐ Calculate, plot errors

UMR species HR species

UMR specialty time series

ACSM average mass spectra plots

Species: Org, NO₃, SO₄, NH₄, CH₃, H₂O, RIE, CE not applied. ☒ Sum HR to UMR ☒ Family colored

UMR species HR species

TOF-ACSM website help

Files Ranges & masks (To-do)

all

NO3CalData

Table of all Table of selected Table of durations

☐ Use buf indices ☒ Hide files

Manually create

Logic:

Clear

Create new Name: myToDo

Overwrite selected Delete selected Delete...

AND OR AND NOT

Range:

0 0

From selection

From to-do

all

Every N: 10

By time of day:

From (hr) 0 to 0

By key wave:

make new (wr based)

Point	t_start
0	11/09/2023 21:31:58
1	11/09/2023 21:32:38
2	11/09/2023 21:33:18
3	11/09/2023 21:33:58
4	11/09/2023 21:34:38
5	12/09/2023 15:08:21
6	12/09/2023 15:09:01
7	12/09/2023 15:09:41
8	12/09/2023 15:10:21
9	12/09/2023 15:11:01
10	12/09/2023 15:12:46
11	12/09/2023 15:13:26
12	12/09/2023 15:14:06
13	12/09/2023 15:14:46
14	12/09/2023 15:15:26

cal_process2_121023

```

312 b = 91.918 ± 5.29
313 •DeletePoints 4,1, ions_avg, ions_std, cpc_counts_avg, cpc_counts_std, flow
314 fit_ions_avg= W_coef[0]+W_coef[1]*x
315 W_coef={25.063,102.45}
316 V_chisq= 3898.29;V_npnts= 4;V_numNaNs= 0;V_numINFs= 0;
317 V_startRow= 0;V_endRow= 3;V_q= 1;V_Rab= -0.81018;V_Pr= 0.997959;
318 V_r2= 0.995923;
319 W_sigma={37.7,4.64}
320 Coefficient values ± one standard deviation
321 a = 25.063 ± 37.7
322 b = 102.45 ± 4.64
323 Saved experiment to: C:\Users\calculs\Documents\ACSM-X\cal_process2_121023
324

```

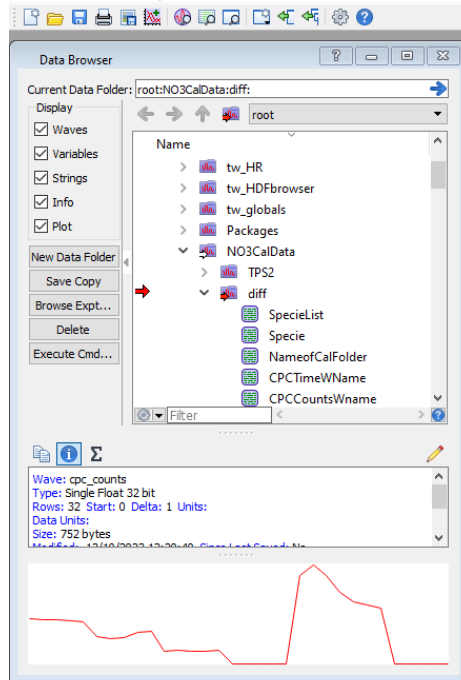
oct. 2023 - 16:47:13:
to workflow: ACSM data processing part 2

oct. 2023 - 16:47:14:
atch table: C:\Users\calculs\Documents\ACSM-
_v3_3_0_ACSM-Software_v3_3_0\WAVES\ACSM_UMRbatch_v01.itx

oct. 2023 - 16:47:14:
R batch table: C:\Users\calculs\Documents\ACSM-
_v3_3_0_ACSM-Software_v3_3_0\WAVES\ACSM_HRbatch_v01.itx

dataset: /TimingData/BufTimes

CSMcal_20230911_143156_diff
CSMcal_20230912_080820_diff
CSMcal_20230912_081244_diff
CSMcal_20230912_081803_diff
CSMcal_20230912_082300_diff



HRCaL

ACSM-X HRCaL.

Name of Cal. Folder: root:NO3CalData:diff:

Specie: select ☐ Use Total Signal ?

☐ CPC Analog Input ? CPC Counts Wave:

CPC Time Wave:

Plot TimeSerie

Cal. Plot & Fit

```
cal_process2_121023
312 b =91.918 ± 5.29
313 DeletePoints 4,1, ions_avg,ions_std,cpc_counts_avg,cpc_counts_std,flo
314 fit_ions_avg= W_coef[0]+W_coef[1]*x
315 W_coef={25.063,102.45}
316 V_chisq= 3898.29;V_npnts= 4;V_numNaNs= 0;V_numINFs= 0;
317 V_startRow= 0;V_endRow= 3;V_q= 1;V_Rab= -0.81018;V_Pr= 0.997959;
318 V_r2= 0.995923;
319 W_sigma={37.7,4.64}
320 Coefficient values ± one standard deviation
321 a =25.063 ± 37.7
322 b =102.45 ± 4.64
323 Saved experiment to: C:\Users\calculs\Documents\ACSM-X\cal_process2_12
324
```

Tofware | cal_process2_121023

File Edit Data Analysis Statistics Macros Windows Panel Misc Tofware Tools Graphs ACSM-X HRCaL Help

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Execute Cmd...

Name

- > tw_HR
- > tw_HDFbrowser
- > tw_globals
- > Packages
- > NO3CalData
 - > TPS2
 - > diff
 - SpeciesList
 - Specie
 - NameofCalFolder
 - CPCTimeWName
 - CPCCountsWName

Filter

Wave: cpc_counts

Type: Single Float 32 bit

Rows: 32 Start: 0 Delta: 1 Units:

Data Units:

Size: 752 bytes

Model: 13/10/2023 13:28:48

ACSM-X HRCaL.

Name of Cal. Folder: root:NO3CalData:diff:

Specie: NO3

Use Total Signal ?

CPC Counts Wave

CPC Time Wave

Plot TimeSerie

Cal. Plot & Fit

Select specie. Only NO3 works so far

Instead of using eg NO+ & NO2+, use total NO3. Disabled so far.

cal_process2_121023

```
312 b = 91.918 ± 5.29
313 •DeletePoints 4,1, ions_avg,ions_std,cpc_counts_avg,cpc_counts_std,flo
314 fit_ions_avg= W_coef[0]+W_coef[1]*x
315 W_coef={25.063,102.45}
316 V_chisq= 3898.29;V_npnts= 4;V_numNaNs= 0;V_numINfs= 0;
317 V_startRow= 0;V_endRow= 3;V_q= 1;V_Rab= -0.81018;V_Pr= 0.997959;
318 V_r2= 0.995923;
319 W_sigma={37.7,4.64}
320 Coefficient values ± one standard deviation
321 a =25.063 ± 37.7
322 b =102.45 ± 4.64
323 Saved experiment to: C:Users:calculs:Documents:ACSM-X:cal_process2_12
324
```

Taper ici pour rechercher

23°C Eclaircies 12:27 13/10/2023

Software | cal_process2_121023

File Edit Data Analysis Statistics Macros Windows Panel Misc Tofware Tools Graphs ACSM-X HRCaL Help

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Name

- tw_HR
- tw_HDFBrowser
- tw_globals
- Packages
- NO3CalData
 - TPS2
 - diff
 - SpecieList
 - Specie
 - NameofCalFolder
 - CPCTimeWName
 - CPCCountsWName

Filter

Wave: cpc_counts

Type: Single Float 32 bit

Rows: 32 Start: 0 Delta: 1 Units:

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Size: 752 bytes

Modif: 12/10/2023 12:28:48

ACSM-X HRCaL.

Name of Cal. Folder: root:NO3CalData:diff:

Specie: NO3

Use Total Signal?

☒ CPC Analog Input? Analog Input Nb: 3

Analog Conv. factor: 1000

Plot TimeSerie

Cal. Plot & Fit

If analog input, just need to set the AI nb used, and conversion factor from V to #

Procedure will get the corresponding wave in TPS2 subfolder

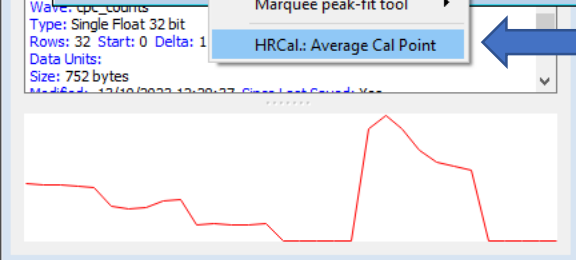
If no AI, CPC counts wave & CPC time wave must be stored in diff folder.

Wave names need to be entered instead.

cal_process2_121023

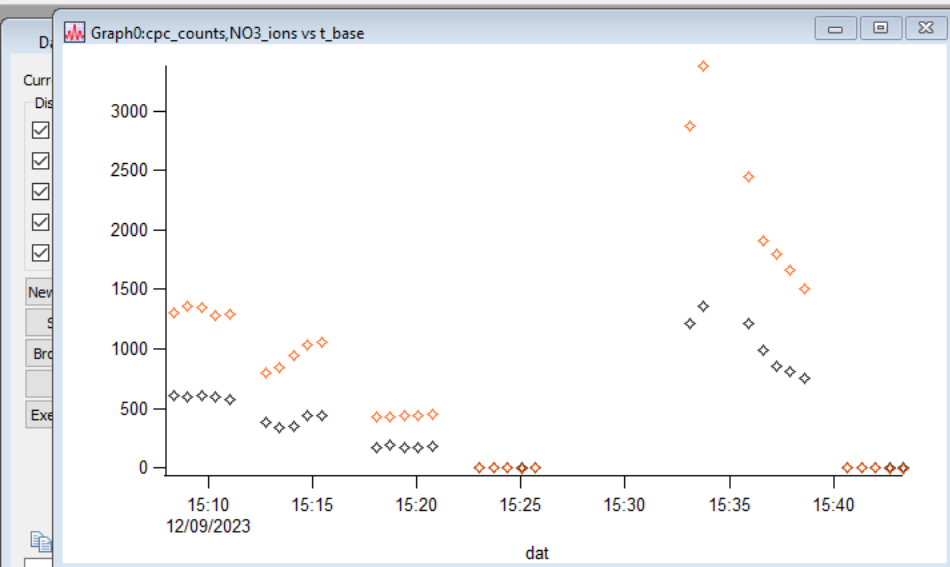
```
312 b = 91.918 ± 5.29
313 •DeletePoints 4,1, ions_avg, ions_std, cpc_counts_avg, cpc_counts_std, flow
314 fit_ions_avg= W_coef[0]+W_coef[1]*x
315 W_coef={25.063,102.45}
316 V_chisq= 3898.29; V_npnts= 4; V_numNaNs= 0; V_numINFs= 0;
317 V_startRow= 0; V_endRow= 3; V_q= 1; V_Rab= -0.81018; V_Pr= 0.997959;
318 V_r2= 0.995923;
319 W_sigma={37.7,4.64}
320 Coefficient values ± one standard deviation
321 a = 25.063 ± 37.7
322 b = 102.45 ± 4.64
323 Saved experiment to: C:\Users\calculs\Documents\ACSM-X\cal_process2_121023
324
```

Windows Taskbar: 23°C Eclaircies 12:28 13/10/2023

[illegible]

Just draw a marquee to select a time window to average

```
cal_process2_121023
312 b =91.918 ± 5.29
313 •DeletePoints 4,1, ions_avg,ions_std,cpc_counts_avg,cpc_counts_std,flow
314 fit_ions_avg= W_coef[0]+W_coef[1]*x
315 W_coef={25.063,102.45}
316 V_chisq= 3898.29;V_npts= 4;V_numNaNs= 0;V_numINFs= 0;
317 V_startRow= 0;V_endRow= 3;V_q= 1;V_Rab= -0.81018;V_Pr= 0.997959;
318 V_r2= 0.995923;
319 W_sigma={37.7,4.64}
320 Coefficient values ± one standard deviation
321 a =25.063 ± 37.7
322 b =102.45 ± 4.64
323 Saved experiment to: C:Users:calculs:Documents:ACSM-X:cal_process2_12
324 |
```



HRCal

ACSM-X HRCal.

Name of Cal. Folder: root:NO3CalData:diff.

Specie: NO3 ☐ Use Total Signal ?

☒ CPC Analog Input ? Analog Input Nb: 3
Analog Conv. factor: 1000

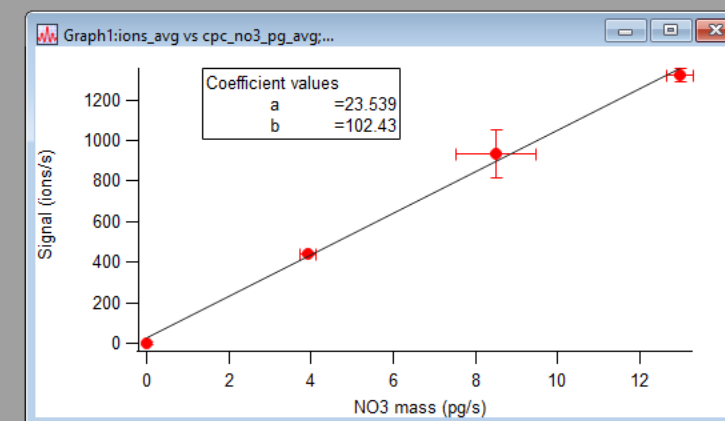
Plot TimeSerie

Cal. Plot & Fit

Table0:ions_avg,ions_std,cpc_counts_avg,cpc_counts_std,flowrate_avg

R0: 1319.8193

Point	ions_avg	ions_std	cpc_counts_avg	cpc_counts_std	flowrate_avg
0	1319.82	35.3214	599	15.1467	1.43633
1	936.048	115.892	391.389	45.4771	1.43893
2	438.97	8.56819	180.278	9.03738	1.44195
3	-1.04219	5.67637	0	0	1.44453
4					



After avg, automatic plot and quick fit

```
cal_process2_121023
325 •AverageCalPoint()
326 •AverageCalPoint()
327 •AverageCalPoint()
328 fit_ions_avg= W_coef[0]+W_coef[1]*x
329 W_coef={23.539,102.43}
330 V_chisq= 3678.88;V_npnts= 4;V_numNaNs= 0;V_numINFs= 0;
331 V_startRow= 0;V_endRow= 3;V_q= 1;V_Rab= -0.793617;V_Pr= 0.998153;
332 V_r2= 0.996309;
333 W_sigma={35.2,4.41}
334 Coefficient values ± one standard deviation
335 a =23.539 ± 35.2
336 b =102.43 ± 4.41
337
```